p8106_hw1_sy3352

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```r
library(glmnet)
library(caret)
library(tidymodels)
library(corrplot)
library(ggplot2)
library(plotmo)
```

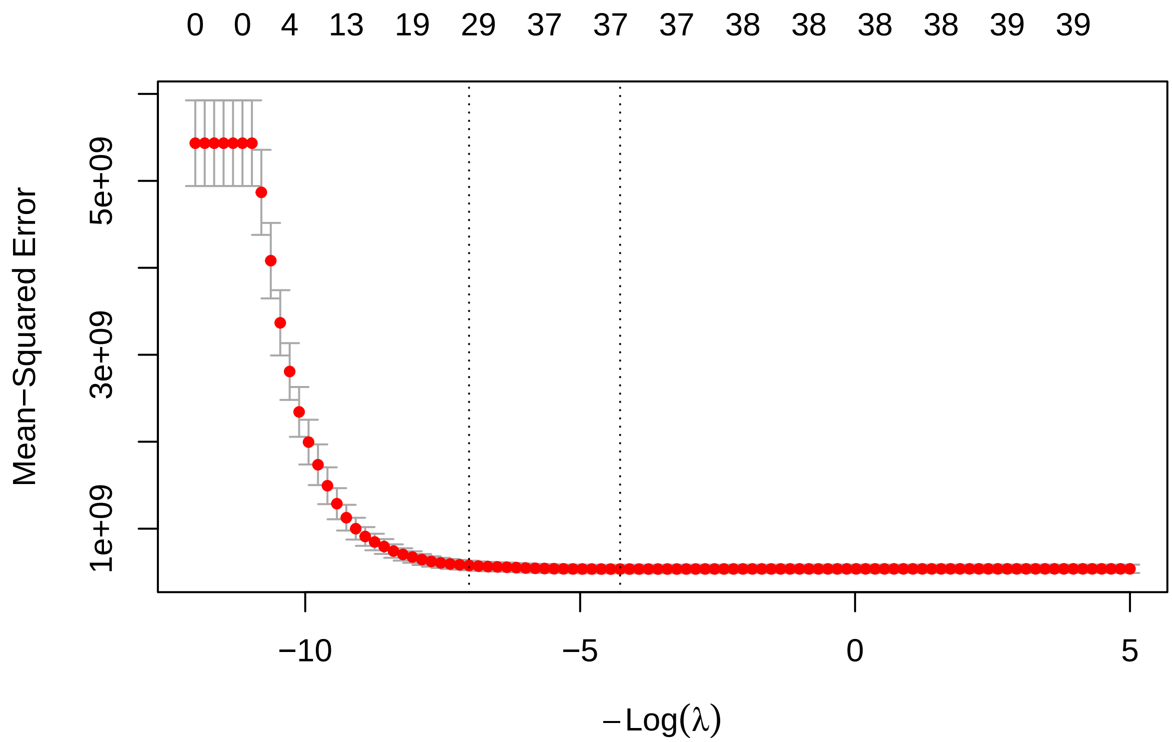
```r
training_data <- read.csv("housing_training.csv") |> na.omit()
testing_data <- read.csv("housing_test.csv") |> na.omit()

x <- model.matrix(Sale_Price ~ ., training_data)[,-1]
y <- training_data[, "Sale_Price"]
```

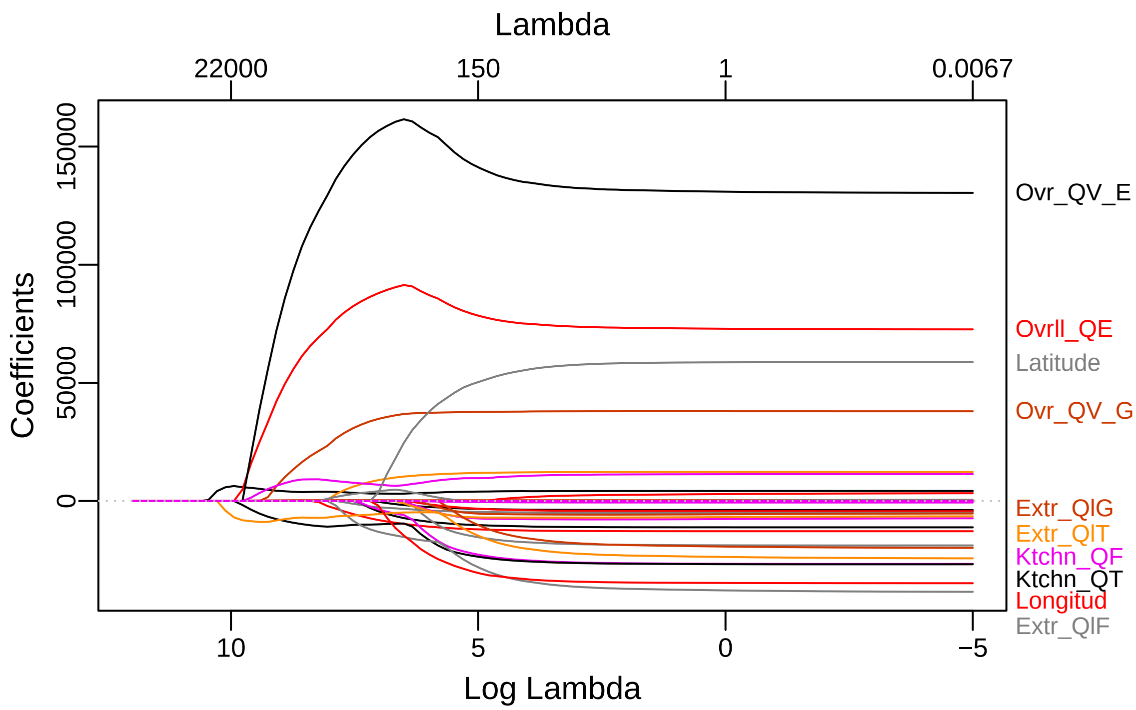
a)

```r
set.seed(2026)
cv.lasso <- cv.glmnet(x, y,
                      alpha = 1,
                      lambda = exp(seq(12, -5, length = 100)))

plot(cv.lasso)
```




```
plot_glmnet(cv.lasso$glmnet.fit)
```




```r
cv.lasso$lambda.1se
```

```
## [1] 1119.013
```

```r
coef_1se <- predict(cv.lasso, s = "lambda.1se", type = "coefficients")
sum(coef_1se != 0) - 1
```

```
## [1] 29
```

```r
lasso_pred <- predict(
  cv.lasso, newx = model.matrix(Sale_Price ~ ., testing_data)[,-1],
             s = "lambda.1se", type = "response")

lasso_test_RMSE = sqrt(mean((lasso_pred - testing_data$Sale_Price)^2))
lasso_test_RMSE
```

```
## [1] 20651.86
```

The selected tuning parameter is 1119.0126581. The test error is 2.0651861×10^4. Using 1SE rule, the number of predictors included is 29.

b)

```r
ctrl1 <- trainControl(method = "cv", number = 10)

set.seed(2026)
enet.fit <- train(Sale_Price ~ .,
                  data = training_data,
                  method = "glmnet",
                  tuneGrid = expand.grid(alpha = seq(0, 1, length = 21),
                                         lambda = exp(seq(12, -5, length = 100))),
                  trControl = ctrl1)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```r
enet.fit$bestTune
```

```
##     alpha   lambda
## 167  0.05 563.0302
```

```r
enet.pred <- predict(enet.fit, newdata = testing_data)
enet_test_RMSE = sqrt(mean((enet.pred - testing_data[, "Sale_Price"])^2))
enet_test_RMSE
```

```
## [1] 20947.88
```

The selected tuning parameter is 0.05, 563.030236835952. The test error is 2.094788×10^4. 1SE is not applicable in elastic net because it has 2 tuning parameters, alpha and lambda. There is no way to apply one standard error of the minimum to 2 parameter. Therefore, 1SE is not applicable here.